MixCT: Mixing Confidential Transactions from Homomorphic Commitment

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Abstract. Mixing protocols serve as a promising solution to the unlinkability in blockchains. They work by hiding one transaction among a set of transactions and enjoy the advantage of high compatibility with the underlying system. However, due to the inherent public addresses of the blockchains built on the account-based model, the unlinkability is highly restricted to non-confidential transactions. In this paper, we propose MixCT, a mixing service for confidential payment systems built from homomorphic commitment in the account-based model. We provide an efficient instantiation of MixCT by the Pedersen commitment and the one-out-of-many proof. The evaluation results show that MixCT introduces a small cost for its users while being highly compatible with the underlying blockchain.

1 Introduction

The decentralized payment system, e.g., blockchain, provides a popular medium of exchange and has attracted a substantial number of applications. Each transaction in a blockchain is confirmed by a public ledger, with the corresponding relationship between the sender and receiver precisely kept. This, however, makes it possible to track the transaction flow and gives rise to the privacy issue. To this point, several privacy-enhanced blockchain systems have been proposed to protect ledger privacy, mainly in two aspects: the unlinkability between transactions and the confidentiality of each transaction's value. In the early blockchain-based cryptocurrencies [14], people use a pseudonym to break the link between a user's on-chain address and the real-world identity. This method, however, has been

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proved to be useless after being thoroughly studied [3,13]. Later, many efforts have been made to provide stronger unlinkability, by mixing multiple transactions and obscuring the relations between the two sets of senders and receivers [1,16,17]. Unlike stand-alone cryptocurrencies such as [4,10,18,19], this approach is highly compatible with the underlying blockchain and needs neither to start up a new blockchain nor to hard-fork an existing one.

The mixing procedure can be implemented with or without a hub. CoinShuffle [16] is a typical non-hub scheme to mix Bitcoin. Users are required to encrypt and decrypt their output addresses in a pre-designed order. To generate CoinJoin transactions in an arbitrary order, DiceMix is proposed in CoinShuffle++ [17]. These two schemes deal with the *non*-confidential transaction where the transaction value is public. ValueShuffle [15], based on DiceMix [17], can mix the confidential transaction by hiding the value using cryptographic commitment. All these *non-hub* schemes suffer from large off-chain communication overhead, e.g., CoinShuffle++ [17] and ValueShuffle [15] require every user to communicate with each of the other participants to generate a CoinJoin transaction. The cost is even more when there are malicious participants. In contrast to the non-hub mixing, in the *hub-based* solutions, a *trustless* tumbler acting as the hub provides the *mixing service*. In this way, neither involved P2P mixing nor participant coordination is required. TumbleBit [11] and A^2L [20] are typical hub-based schemes that also prevent the tumblers from linking users during mixing. However, most existing hub-based mixing service works for *non*-confidential blockchain only. Since the participants' addresses are public, anyone can link two participants by matching the equal values on the input and output sides. So only *fixed* and equal value mixing is supported [11,12,20]. As far as we know, the only confidential payment system that supports *arbitrary* transaction value mixing is [15], a non-hub mixing in the unspent-transaction-output (UTXO) model.

Unlike the UTXO model built from Bitcoin, the account-based model introduced by Ethereum supports smart contracts for real-world applications. Møbius [12] uses a smart contract as a tumbler for the first time, but it does not support confidential mixing. Most recently, people have proposed several confidential payment systems in the account-based model, e.g., basic Zether [5] and PGC [7]. Unfortunately, neither of them supports unlinkability. Though [5] and [8] use sophisticated ZK-proofs and many-out-of-many proofs to allow anonymous transactions, the resulting schemes are not compatible with the original Zether anymore. To the best of our knowledge, providing a mixing service for the confidential payment system in the account-based model is still open.

Table 1. Comparison with previous mixing solutions (n: the number of mixing users)

Type	Scheme	# Off-Chain Messages	# Transactions /User	Model	Confidential Mixing	Payment Value	DoS Resistance
Non-hub	CoinShuffle [16]	$\mathcal{O}(n)$	1	UTXO	✗	Fixed	N/A
	CoinShuffle++ [17]	$\mathcal{O}(n)$	1	UTXO	✗	Fixed	N/A
	ValueShuffle [15]	$\mathcal{O}(n)$	1	UTXO	✓	Arbitrary	N/A
Hub-based	TumbleBit [11]	12	4	U or A	✗	Fixed	✗
	A^2L [20]	9	4	U or A	✗	Fixed	✓
	Möbius [12]	2	2	Account	✗	Fixed	✓
	Ours	2	2	Account	✓	Arbitrary	✓

This work addresses this problem with MixCT, which provides mixing services for the confidential payment systems in the account-based model. Specifically, for any confidential payment system built from the homomorphic commitment in the account-based model, MixCT can provide the mixing service with a trustless tumbler, without changing the transaction format. We provide the security goals of MixCT and analyze them. Furthermore, we instantiate MixCT with Pedersen commitment and one-out-of-many proof on Ethereum. The evaluation results show that our MixCT is cheap compared with other unlinkability solutions for confidential payment systems. A comparison between MixCT and the state-of-the-art mixing approaches is given in Table 1.

2 MixCT Design

MixCT works on top of a confidential payment system built in the account-based model and provides a mixing service for the underlying systems by a *trustless* tumbler. Roughly, the tumbler works as an unlikable payment hub. Each mixing service user has two accounts, called the sender and the receiver. Senders escrow their coins to the tumbler, and receivers redeem coins from it. For safety, the tumbler cannot violate the unlinkability between the escrow and redeeming transactions, "print money", or steal money from users. We use a smart contract to realize the tumbler.

Specifically, in MixCT, the tumbler divides the time into epochs, and one epoch consists of two phases: *escrow* and *redeeming*. Besides, the tumbler *publicly* maintains its $\mathsf{state} = (\mathsf{Epool}, \mathsf{Rpool})$. The workflow is shown in Fig. 1.

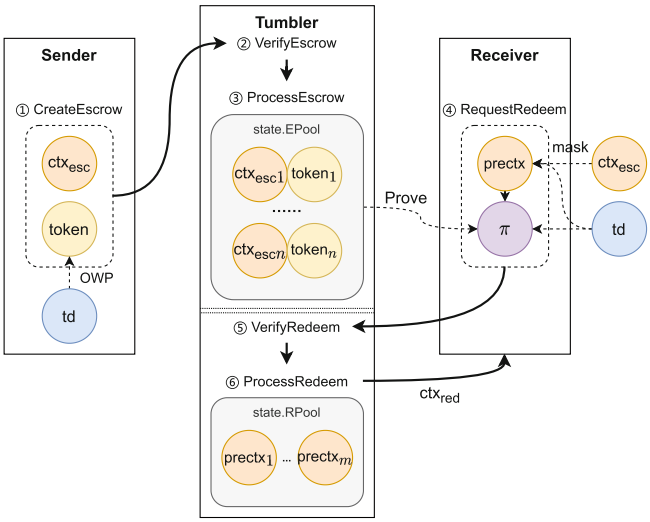

Fig. 1. Overview of MixCT: this picture explains one payment between one pair of sender and receiver in one epoch. Black solid lines show the interactions among the sender, the tumbler, and the receiver. Black dashed lines show the data dependencies

Escrow Phase. The sender creates an escrow transaction by generating a confidential transaction $\mathsf{ctx}_{\mathrm{esc}}$ to the tumbler. In addition, the sender generates a secret random factor td and hides it by utilizing a one-way-permutation (OWP) to generate a *public* token (① in Fig. 1). The tumbler verifies the escrow transaction pair $(\mathsf{ctx}_{\mathrm{esc}}, \mathsf{token})$ by validating $\mathsf{ctx}_{\mathrm{esc}}$ via the underlying chain and confirming the uniqueness of the token (② in Fig. 1). If both checks pass, the tumbler records the new received escrow transaction $(\mathsf{ctx}_{\mathrm{esc}}, \mathsf{token})$ and publicly updates its state, to finish the processing of the escrow (③ in Fig. 1).

Redeeming Phase. The receiver, who gets the token's preimage td of an escrow transaction $(\mathsf{ctx}_{\mathrm{esc}}, \mathsf{token})$ in the state, launches the request of redeeming by generating a confidential pre-redeeming transaction prectx and a publicly verifiable redeeming proof π to be sent to the tumbler (④ in Fig. 1). In detail, the receiver uses td as the additional random factor and the homomorphic property of the confidential transaction to mask $\mathsf{ctx}_{\mathrm{esc}}$. In this way, prectx hides the same value as $\mathsf{ctx}_{\mathrm{esc}}$. π is used to prove that prectx is derived from an existing $(\mathsf{ctx}_{\mathrm{esc}}, \mathsf{token})$ in state. The tumbler verifies the redeeming request by confirming the format, the uniqueness of prectx, and the validity of the proof π (⑤ in Fig. 1). If all checks pass, the tumbler finishes the processing of redeeming by running on prectx to generate the complete redeeming transaction $\mathsf{ctx}_{\mathrm{red}}$, and recording prectx to update its state (⑥ in Fig. 1).

In summary, in each epoch, a one-to-one transparent permutation is formed among the redeeming and escrow transactions. One escrow can create only one successful redeeming, since each $(\mathsf{ctx}_{\mathrm{esc}}, \mathsf{token})$ determines the corresponding

prectx. Due to the hard-to-invert property of the one-way-permutation, the relation between $(\mathsf{ctx}_{\mathrm{esc}}, \mathsf{token})$ and its prectx is invisible to anyone (including the tumbler) apart from the two owners of td. Moreover, MixCT is fully compatible with the underlying confidential blockchain, because $\mathsf{ctx}_{\mathrm{red}}$ and $\mathsf{ctx}_{\mathrm{esc}}$ have the same format.

3 Security Goals

Previous works [12,15,16] have discussed the security goals for the mixing service. We extend them to the account-based model and the confidential transaction setting. The formal definitions and analysis are provided in the full version [9].

- **Theft prevention.** The achieved goal is two-fold. (1) For an accepted escrow transaction $(\mathsf{ctx}_{\mathrm{esc}}, \mathsf{token})$ generated by a sender, nobody can redeem it without the transaction's td from the sender. (2) Nobody can redeem "out of nothing". That is, nobody can generate a valid redeeming request (prectx, π) without a corresponding escrow transaction in state.
- **Transaction balance.** Since $\mathsf{ctx}_{\mathrm{esc}}$ and prectx share the same transaction value, every escrowed transaction can be redeemed with the same value as the corresponding sender has escrowed.
- **Double-spend prevention.** The one-to-one relation between the redeeming and escrow transactions ensures that every escrowed transaction cannot be redeemed more than once.
- **Unlinkability.** Since the above-mentioned one-to-one permutation is invisible to anyone excluding the actual sender and receiver, the tumbler cannot link a redeeming transaction with an escrow transaction, among a sequence of escrow and redeeming.
- **Confidentiality.** Since prectx and $\mathsf{ctx}_{\mathrm{red}}$ are confidential, i.e., protected by the underlying homomorphic commitment, no information about the transaction value is leaked during the mixing procedure, which means that the introduction of MixCT will not violate the underlying confidentiality.
- **DoS resistance.** Since the sender pays the tumbler first to get service, the expenses incurred launching the DoS attacks has been raised. Besides, every honest party can always terminate without losing coins.

4 Implementation and Evaluation

To show the feasibility of MixCT, we implement the underlying confidential payment with the well-known Pedersen homomorphic commitment. We generate the public token from td, the one-out-of-many proof to form the proof π of prectx, based on the assumption of the hardness of the discrete logarithm problem. All the generation is carried out without a trusted setup. Then we implement MixCT with Ethereum smart contract. To evaluate the performance of MixCT, we vary the number of the mixing service users and record the costs for escrow and

redeeming accordingly. The costs of escrow lie in the generation of the token. In the redeeming, the costs result from the user's off-line generation of the one-out-of-many proof, and the tumbler's online verification and processing.

The evaluation results of MixCT are given in Table 2. As can be seen, the escrow costs are fixed for all, regardless of the number of the users. The redeeming costs increase from 496 k Gas to 3085 k Gas as the escrow users' number changes from 4 to 64, which is much lower than other sophisticated approaches [5,8]. Moreover, to generate/validate the proof of a redeeming transaction with no more than 64 mixing users, the time cost is less than 1.5/0.5 s.

Table 2. The evaluation results of MixCT

#users	Escrow		Redeem			
	Size/bytes	Gas/units	Size/bytes	Prove Time/ms	Verify Time/ms	Gas/units
$n = 4$	132	128,556	2,500	173	93	496,163
$n = 8$			2,948	271	133	714,341
$n = 16$			3,396	451	185	1,080,093
$n = 32$			3,844	758	288	1,755,951
$n = 64$			4,292	1,466	462	3,085,444
n	$\mathcal{O}(1)$		$\mathcal{O}(\log n)$	$\mathcal{O}(n\log n)$		

Our evaluation benchmarks are collected on a laptop with a 2.9 GHz AMD R7-4800H processor. We implement our design on Truffle Suite [2], We use Solidity to accomplish the smart contract of the tumbler running on top of the Ethereum virtual machine (EVM) and use JavaScript to implement the mixing users. Particularly, we use the construction of one-out-of-many proof from [6]. All code has been open-sourced[1].

5 Conclusion

In this work, we have proposed a confidential mixing service scheme in the account-based model. The proposed scheme, MixCT, allows us to mix confidential transactions containing arbitrary transaction values. Experimental evaluation shows that MixCT could be instantiated by lightweight cryptographic tools without a trusted setup.

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